

Answer Part 1: Git Use with Engineering Workflow

1. GitHub URLs

GitHub Repository: <https://github.com/chayongchaya/toktickit>

GitHub Project: <https://github.com/users/chayongchaya/projects/16>

GitHub Issues:

Issue #1: <https://github.com/chayongchaya/toktickit/issues/1>

Issue #2: <https://github.com/chayongchaya/toktickit/issues/2>

Issue #3: <https://github.com/chayongchaya/toktickit/issues/3>

Issue #4: <https://github.com/chayongchaya/toktickit/issues/4>

Pull Requests:

PR #1 - Feature branch: <https://github.com/chayongchaya/toktickit/pull/5>

PR #2 - Feature branch: <https://github.com/chayongchaya/toktickit/pull/6>

PR #3 - Feature branch: <https://github.com/chayongchaya/toktickit/pull/7>

PR #4 - Feature branch: <https://github.com/chayongchaya/toktickit/pull/8>

PR - docs: <https://github.com/chayongchaya/toktickit/pull/9>

PR - main: <https://github.com/chayongchaya/toktickit/pull/10>

2. GitHub Project / Kanban

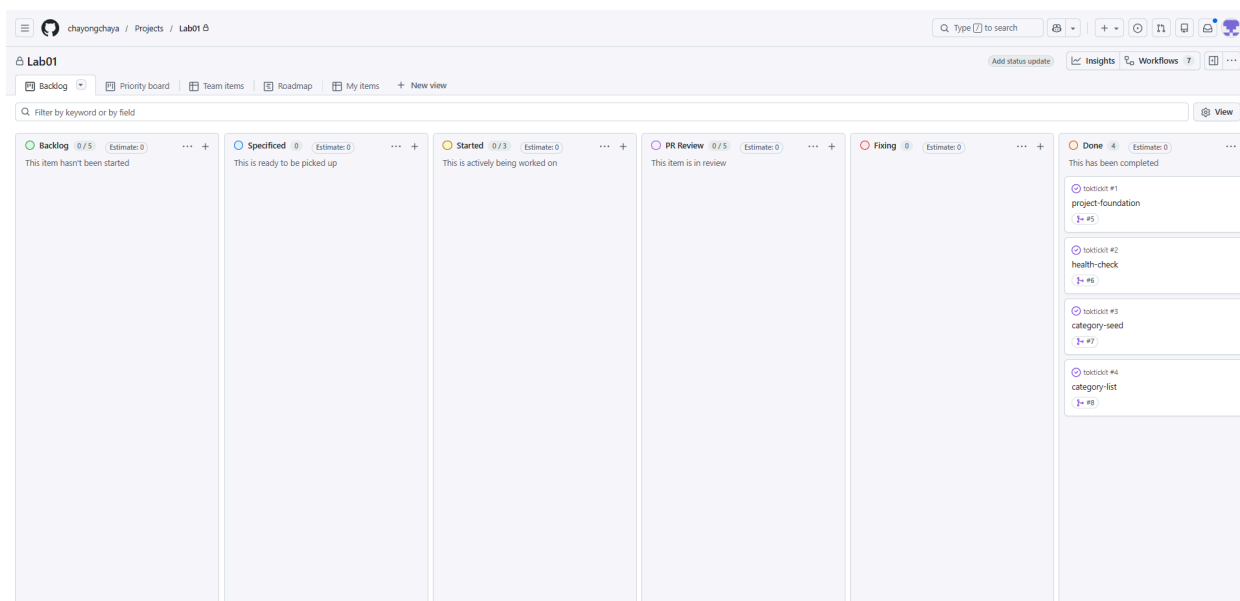


figure 1 : Part 1 Github Kanban board with done issue

3. Git Workflow / Commit History

Commits

 main	 All users	 All time
 Commits on Aug 15, 2026		
Update document  Kulchaya Paipinij committed 2 minutes ago 3837f74  		
Update README  Kulchaya Paipinij committed 31 minutes ago e572d86  		
Merge pull request #10 from chayongchaya/lab1-staging  chayanitkunt authored 2 hours ago 70681e4   Verified		
Merge pull request #9 from chayongchaya/feature/5-update-docs  chayanitkunt authored 2 hours ago 66dc7cd   Verified		
docs: update documentation  Kulchaya Paipinij committed 2 hours ago 8dccc98  		
 Commits on Aug 14, 2026		
Merge pull request #8 from chayongchaya/feature/4-categories-api  chayanitkunt authored 2 hours ago e11464f   Verified		
feat: implement GET /api/categories and frontend display  Kulchaya Paipinij committed 2 hours ago 54fcd6d  		
Merge pull request #7 from chayongchaya/feature/3-category-seed  chayanitkunt authored 3 hours ago fc02018   Verified		
feat: add category model, migration, and seed script  Kulchaya Paipinij committed 3 hours ago 7c320cf  		
Merge pull request #6 from chayongchaya/feature/2-health-check  chayanitkunt authored 3 hours ago 0f36287   Verified		
feat: implement GET /api/health endpoint and pass health test  Kulchaya Paipinij committed 3 hours ago a43ce4a  		
Merge pull request #5 from chayongchaya/feature/1-project-foundation  chayanitkunt authored 3 hours ago a0c744c   Verified		
chore: setup project foundation, dependencies, and gitignore  Kulchaya Paipinij committed 4 hours ago 8fef04b  		
first commit  Kulchaya Paipinij committed 5 hours ago d254cb4  		

figure 2 : Part 1 Commit History

4. Repository Directory Structure

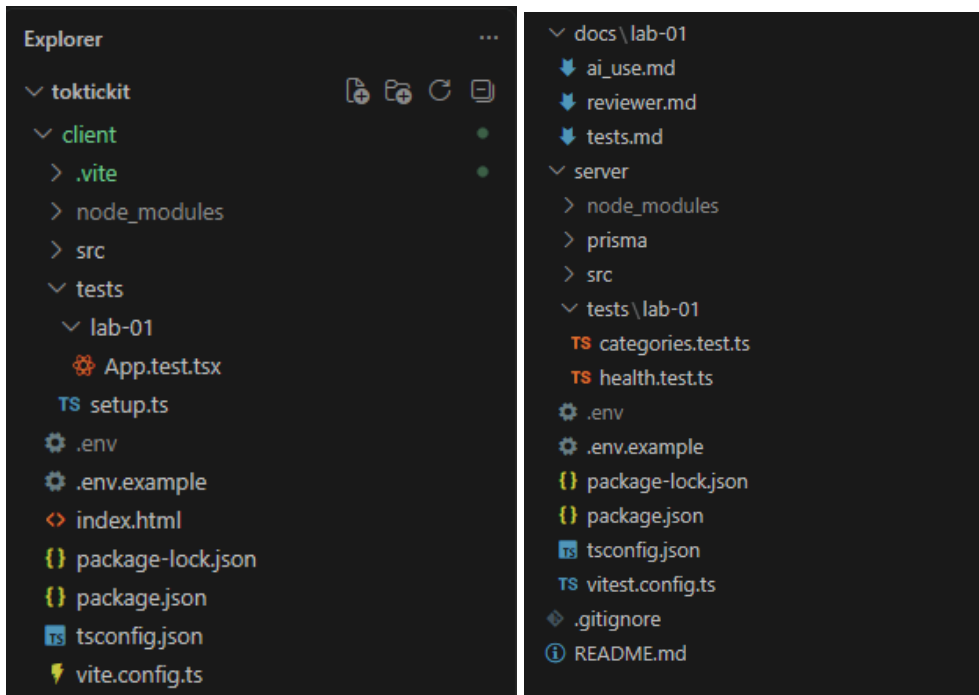


figure 3 : Part 1 Repository Directory Structure (Part1)

figure 4 : Part 1 Repository Directory Structure (Part 2)

5. README.md

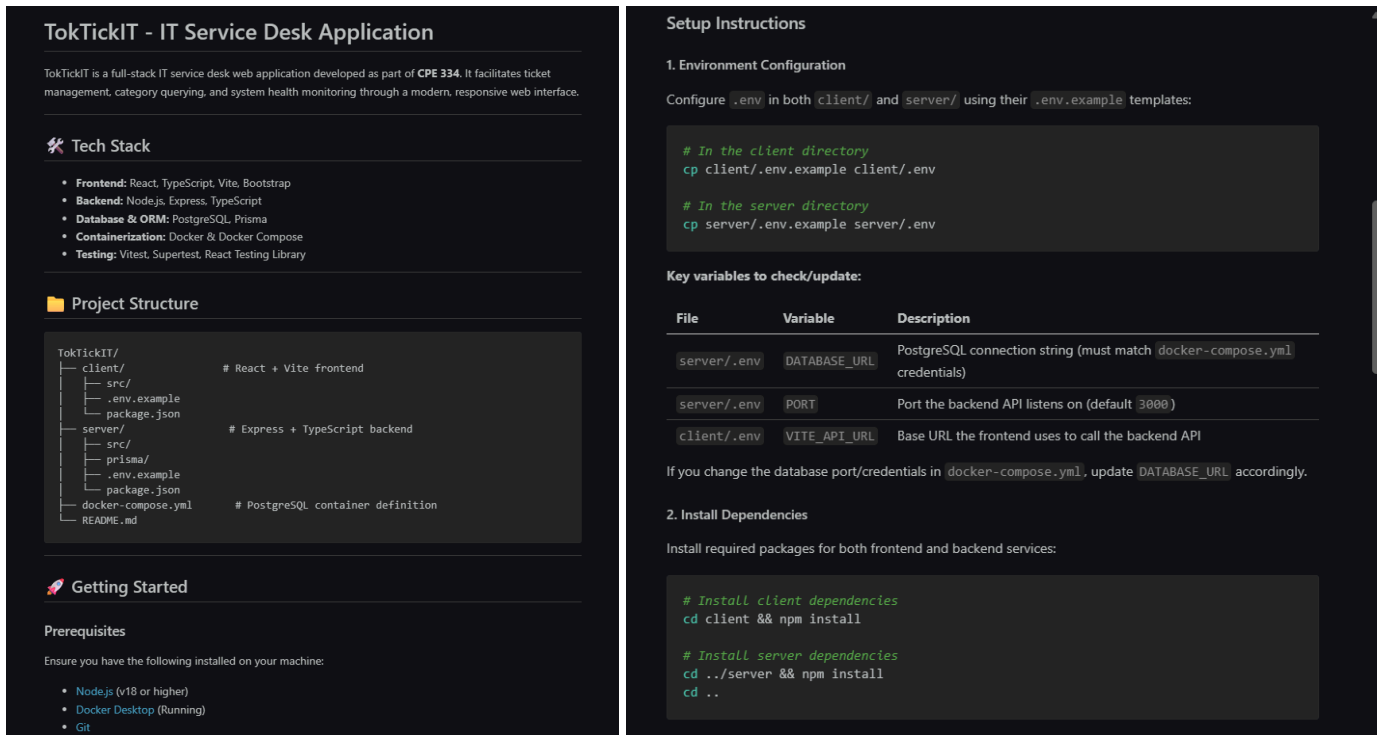


figure 5 : Part 1 Readme.md file (Part1)

figure 6 : Part 1 Readme.md file(Part 2)

3. Database Initialization (Docker & Prisma)

Start the PostgreSQL container and apply schema migrations with seed data:

```
# Start PostgreSQL container in background (from root directory)
docker compose up -d

# Navigate to server and apply schema migrations
cd server
npx prisma migrate dev

# Seed initial categories into the database
npx prisma db seed
cd ..
```

Running the Application

Open two separate terminals to run both services concurrently:

1. Start Server (Backend API)

```
cd server
npm run dev
```

Server runs on `http://localhost:3000`

2. Start Client (Frontend Web App)

```
cd client
npm run dev
```

Client runs on `http://localhost:5173` by default. If port `5173` is already in use, Vite automatically switches to the next available port (e.g. `5174`) — check your terminal output for the exact URL.

Running Automated Tests

Backend Test Suite

Executes integration tests for API routes (`/health` and `/api/categories`):

```
cd server
npm test
```

Frontend Test Suite

Executes component tests and Mock UI verification:

```
cd client
npm test
```

API Reference (Lab 1)

Method	Endpoint	Description	Expected Status
GET	<code>/health</code>	Health check endpoint returning <code>{ status: "ok" }</code>	200 OK
GET	<code>/api/categories</code>	Returns an array of ticket categories ordered by ID	200 OK

Troubleshooting

Problem	Possible Cause	Solution
<code>docker compose up -d</code> fails	Port <code>5432</code> already in use by another PostgreSQL instance	Stop the conflicting service, or change the exposed port in <code>docker-compose.yml</code> and update <code>DATABASE_URL</code>
<code>npx prisma migrate dev</code> fails to connect	Docker container not fully started yet	Wait a few seconds and retry, or check <code>docker ps</code> to confirm the container is running
Client can't reach the API	<code>VITE_API_URL</code> doesn't match the server's actual port	Verify <code>server/.env</code> <code>PORT</code> matches <code>client/.env</code> <code>VITE_API_URL</code>
<code>npm install</code> fails	Nodejs version too old	Confirm <code>node -v</code> is v18 or higher

Contributors

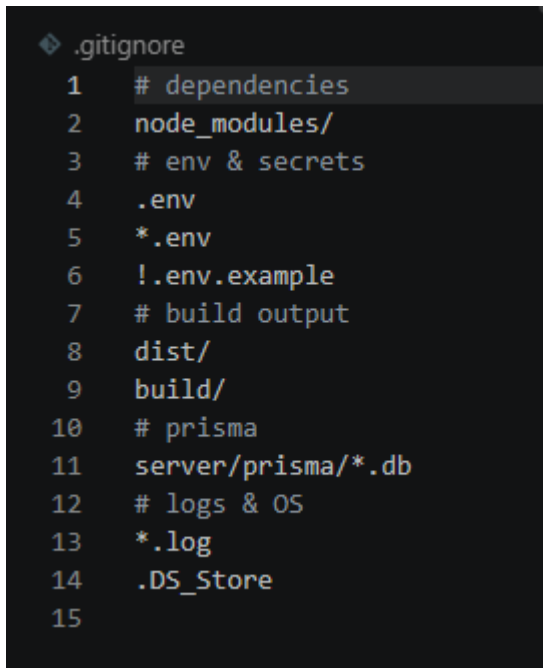
- Student Name: Kulchaya Paipinij
- Student ID: 67070503406
- Course: CPE 334

figure 7 : Part 1 Readme.md file (Part3)

figure 8 : Part 1 Readme.md file(Part 4)

figure 9 : Part 1 Readme.md file(Part 5)

6. .gitignore

A screenshot of a code editor showing the contents of a .gitignore file. The file is titled ".gitignore" with a diamond icon. The content is as follows:

```
1  # dependencies
2  node_modules/
3  # env & secrets
4  .env
5  *.env
6  !.env.example
7  # build output
8  dist/
9  build/
10 # prisma
11 server/prisma/*.db
12 # logs & OS
13 *.log
14 .DS_Store
15
```

figure 10 : Part 1 .gitignore file

7. Peer Review Evidence

7.1 Reviewer Information

Reviewer Name: Chayanit Kuntanarumitkul

Student ID: 67070503408

GitHub Username: chayanitkunt

7.2 Peer Review of My Pull Request

chore: setup project foundation #5

Merged chayankunt merged 1 commit into lab1-staging from feature/1-project-foundation 2 hours ago

Conversation 2 Commits 1 Checks 0 Files changed 29

chayongchaya commented 3 hours ago

Closes #1

chore: setup project foundation, dependencies, and gitignore 8f8f840

chayongchaya requested a review from chayankunt 3 hours ago

chayongchaya linked an issue 3 hours ago that may be closed by this pull request
project-foundation #1

chayankunt approved these changes 2 hours ago View reviewed changes

chayankunt left a comment

This code is good! keep going.

chayongchaya commented 2 hours ago Author

Thanks for suggestion.

chayankunt merged commit abc744c into lab1-staging 2 hours ago Revert

chayongchaya mentioned this pull request 1 hour ago
docs: update documentation #9

Pull request successfully merged and closed
You're all set — the feature/1-project-foundation branch can be safely deleted. Delete branch

feat: implement health check endpoint #6

Merged chayankunt merged 1 commit into lab1-staging from feature/2-health-check 2 hours ago

Conversation 2 Commits 1 Checks 0 Files changed 1

chayongchaya commented 2 hours ago

Closes #2

feat: implement GET /api/health endpoint and pass health test a43ce4a

chayongchaya requested a review from chayankunt 2 hours ago

chayongchaya linked an issue 2 hours ago that may be closed by this pull request
health-check #2

chayankunt approved these changes 2 hours ago View reviewed changes

chayankunt left a comment

Great Job! localhost3000/api/health result should be fine now.

chayongchaya commented 2 hours ago Author

Thank you for helping me

chayankunt merged commit e7a62a7 into lab1-staging 2 hours ago Revert

Pull request successfully merged and closed
You're all set — the feature/2-health-check branch can be safely deleted. Delete branch

feat: create and seed categories #7

Merged chayankunt merged 1 commit into lab1-staging from feature/3-category-seed 1 hour ago

Conversation 2 Commits 1 Checks 0 Files changed 4

chayongchaya commented 1 hour ago

Closes #3

feat: add category model, migration, and seed script 7c328cf

chayongchaya requested a review from chayankunt 1 hour ago

chayongchaya linked an issue 1 hour ago that may be closed by this pull request
category-seed #3

chayankunt approved these changes 1 hour ago View reviewed changes

chayankunt left a comment

You should be able to see the prisma now. Nice job!

chayongchaya commented 1 hour ago Author

Thanks for your kindness.

chayankunt merged commit e6a2a2a into lab1-staging 1 hour ago Revert

Pull request successfully merged and closed
You're all set — the feature/3-category-seed branch can be safely deleted. Delete branch

feat: implement categories API and UI #8

Merged chayankunt merged 1 commit into lab1-staging from feature/4-categories-api 1 hour ago

Conversation 2 Commits 1 Checks 0 Files changed 5

chayongchaya commented 1 hour ago

Closes #4

feat: implement GET /api/categories and frontend display 54fcd86

chayongchaya requested a review from chayankunt 1 hour ago

chayongchaya linked an issue 1 hour ago that may be closed by this pull request
category-list #4

chayankunt approved these changes 1 hour ago View reviewed changes

chayankunt left a comment

great!

chayongchaya commented 1 hour ago Author

Thanks for supporting me!

chayankunt merged commit e11a64f into lab1-staging 1 hour ago Revert

Pull request successfully merged and closed
You're all set — the feature/4-categories-api branch can be safely deleted. Delete branch

figure 11 : Part 1 Peer Review Evidence by my partner (Part1)

figure 12 : Part 1 Peer Review Evidence by my partner (Part2)

figure 13 : Part 1 Peer Review Evidence by my partner (Part3)

figure 14 : Part 1 Peer Review Evidence by my partner (Part4)

docs: update documentation #9

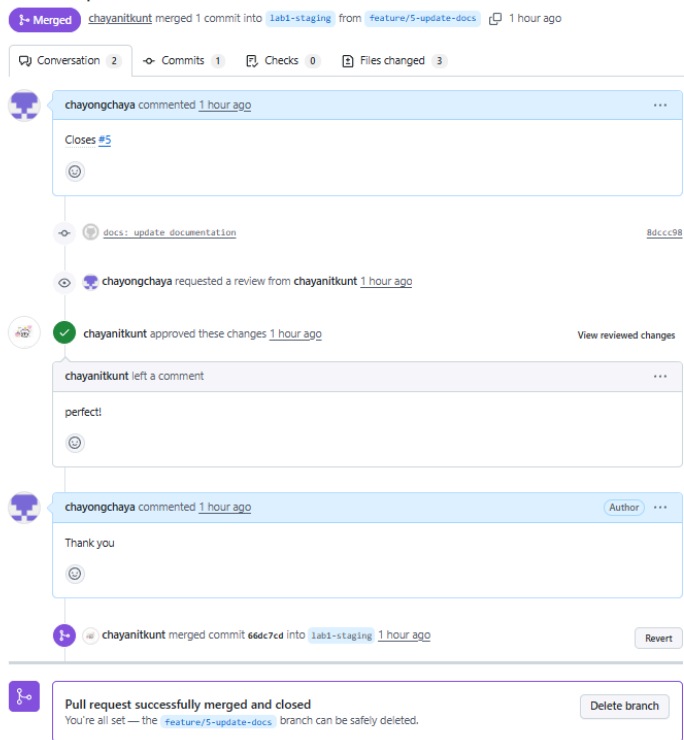


figure 15 : Part 1 Peer Review Evidence by my partner (Part5)

release: merge lab1-staging to main #10

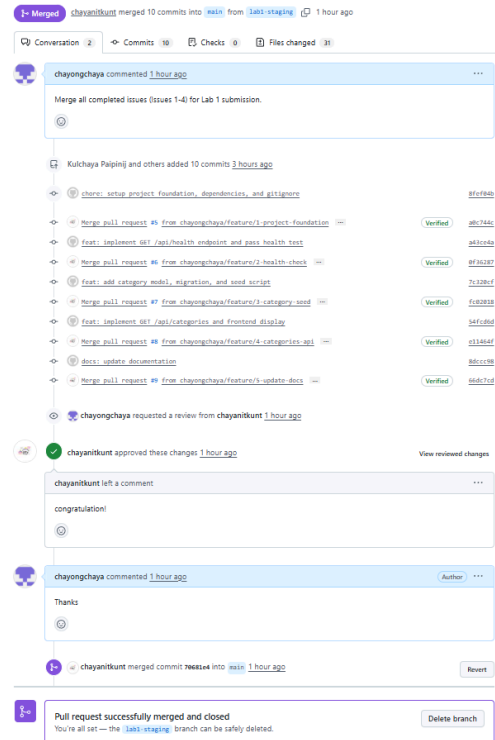


figure 16 : Part 1 Peer Review Evidence by my partner (Part6)

7.3 Peer Review Given to My Partner

feat: complete Lab 1 project foundation #5

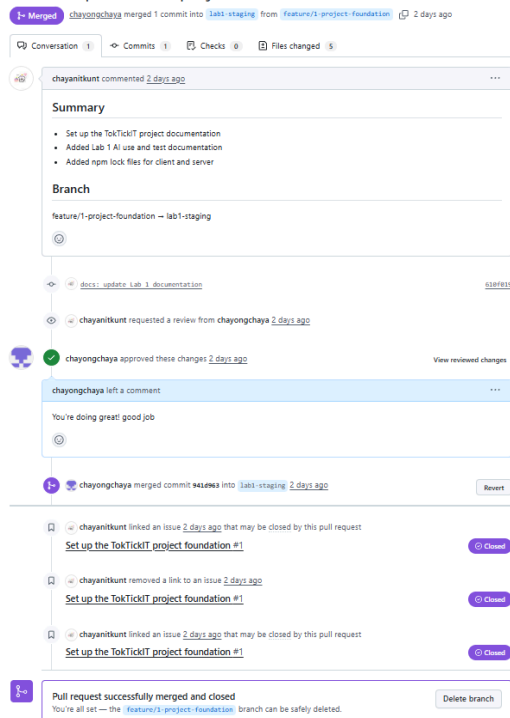


figure 17 : Part 1 Peer Review Evidence given my partner (Part1)

feat: add /api/health endpoint #6

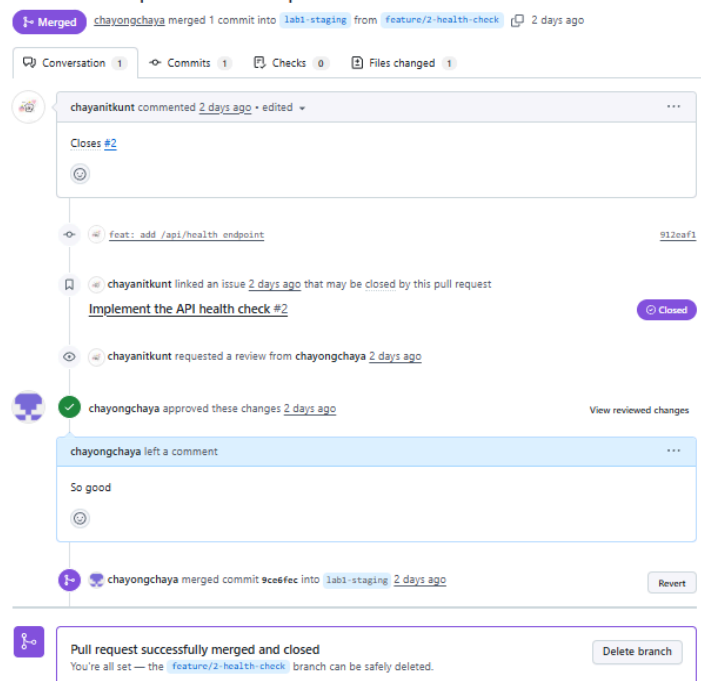


figure 18 : Part 1 Peer Review Evidence given my partner (Part2)

feat: create and seed IT request categories #7

Merged chayongchaya merged 1 commit into lab1-staging from feature/3-category-seed 12 hours ago

Conversation 2 Commits 1 Checks 0 Files changed 4

chayanitkunt commented yesterday

Closes #3

feat: create and seed categories 3352feb

chayanitkunt requested a review from chayongchaya yesterday

chayanitkunt linked an issue yesterday that may be closed by this pull request

Create and seed IT request categories #3

chayongchaya approved these changes 12 hours ago

chayongchaya left a comment

It's great! Easy to look.

chayanitkunt commented 12 hours ago

It's great! Easy to look.
thank you so much

chayongchaya merged commit 66a8ee into lab1-staging 12 hours ago

Pull request successfully merged and closed
You're all set — the feature/3-category-seed branch can be safely deleted.

Delete branch

feat: complete category list feature and unit tests #8

Merged chayongchaya merged 1 commit into lab1-staging from feature/4-category-list 11 hours ago

Conversation 2 Commits 1 Checks 0 Files changed 5

chayanitkunt commented 11 hours ago

Closes #4

feat: complete category list feature and unit tests 5883686

chayanitkunt requested a review from chayongchaya 11 hours ago

chayanitkunt linked an issue 11 hours ago that may be closed by this pull request

Display the IT request category list #4

chayongchaya approved these changes 11 hours ago

chayongchaya left a comment

perfect!

chayanitkunt commented 11 hours ago

Thanks!

chayongchaya merged commit 5ef9ec into lab1-staging 11 hours ago

Pull request successfully merged and closed
You're all set — the feature/4-category-list branch can be safely deleted.

Delete branch

docs: complete lab 1 documentation #9

Merged chayongchaya merged 1 commit into lab1-staging from feature/lab1-documentation 10 hours ago

Conversation 2 Commits 1 Checks 0 Files changed 4

chayanitkunt commented 10 hours ago

Summary

- Complete Lab 1 README setup and running instructions
- Complete AI use and reflection documentation
- Complete peer review record
- Complete test plan and evidence
- Document backend and frontend test results

Testing

- Backend: 2 tests passed
- Frontend: 3 tests passed

Related

Closes #1

docs: complete lab 1 documentation 7899440

chayanitkunt requested a review from chayongchaya 10 hours ago

chayongchaya approved these changes 10 hours ago

chayongchaya left a comment

so good

chayanitkunt commented 10 hours ago

thanks!

chayongchaya merged commit 3f66ac into lab1-staging 10 hours ago

Pull request successfully merged and closed
You're all set — the feature/lab1-documentation branch can be safely deleted.

Delete branch

chore: release lab-01 from staging to main #10

Merged chayongchaya merged 10 commits into main from lab1-staging 1 hour ago

Conversation 2 Commits 10 Checks 0 Files changed 16

chayanitkunt commented 1 hour ago

Summary

- Complete Lab 1 ToxTicket full stack vertical slice
- Complete ToxTicket Get up the ToxTicket project foundation #1
- Add health check and category API
- Add React UI with loading, success, and error states
- Add automated backend and frontend tests
- Complete Lab 1 documentation

Testing

- Backend: 2 tests passed
- Frontend: 3 tests passed

Related

Closes #1

chayanitkunt and others added 10 commits 2 days ago

- docs: update lab 1 documentation 1189951
- feat: add /api/health endpoint 814802
- feat: add /api/health endpoint 813a7f1
- feat: pull request #6 from chayanitkunt/feature/3-health-check 5a0f4ec
- feat: create and seed categories 3352feb
- feat: pull request #7 from chayanitkunt/feature/3-category-seed 58a6a0e
- feat: complete category list feature and unit tests 5883686
- feat: pull request #8 from chayanitkunt/feature/4-category-list 50f4d0c
- docs: complete lab 1 documentation 7899440
- feat: pull request #9 from chayanitkunt/feature/lab1-documentation 3f66ac

chayanitkunt requested a review from chayongchaya 1 hour ago

chayongchaya commented 1 hour ago

Congrat!

chayanitkunt commented 1 hour ago

Thank you for your support so far!

chayongchaya merged commit 51a4de into main 1 hour ago

Pull request successfully merged and closed
You're all set — the lab1-staging branch can be safely deleted.

Delete branch

figure 19 : Part 1 Peer Review Evidence by my partner (Part3)

figure 20 : Part 1 Peer Review Evidence by my partner (Part4)

figure 21 : Part 1 Peer Review Evidence by my partner (Part5)

figure 22 : Part 1 Peer Review Evidence by my partner (Part6)

8. docs/lab-01/reviewer.md

Lab 1 — Peer Review Record (fill this in)

Author: Kulchaya PApinij — 67070503406 — GitHub: @chayongchaya Peer reviewer: Chayanit Kuntanarumitkul — 67070503408 — GitHub: @chayanitkunt

Pull Requests I authored (reviewed by my partner)

PR	Branch	Reviewer verdict
#1	feature/1-project-foundation	Approved
#2	feature/2-health-check	Approved
#3	feature/3-category-seed	Approved
#4	feature/4-category-list	Approved

Reviewer comment I received: "great!" on PR #4 (along with "This code is good! keep going." on PR #1, "Great Job! localhost3000/api/health result should be fine now." on PR #2, and "You should be able to see the prisma now. Nice job!" on PR #3) How I responded: "Thanks for helping me." and verified that all test suites passed before merging into lab1-staging.

Pull Requests I reviewed for my partner

My comment: "perfect!" on PR #4 (along with "You're doing great! good job" on PR #1, "So good" on PR #2, and "It's great! Easy to look." on PR #3) Partner's response: "Thanks!" and proceeded to merge into lab1-staging.

figure 23 : Part 1 reviewer.md

Answer Part 2: Tests

1. Test Evidence

```
PS C:\Users\User\Downloads\toktickit> git status
● On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\User\Downloads\toktickit> cd server
PS C:\Users\User\Downloads\toktickit\server> npm test

> toktickit-server@1.0.0 test
> vitest run

 RUN  v2.1.9 C:/Users/User/Downloads/toktickit/server

 ✓ tests/lab-01/categories.test.ts (1)
 ✓ tests/lab-01/health.test.ts (1)

Test Files  2 passed (2)
Tests       2 passed (2)
Start at    02:45:33
Duration    681ms (transform 86ms, setup 0ms, collect 460ms, tests 82ms, environment 0ms, prepare 312ms)
```

figure 24 : Part 2 test evidence of server

```
● PS C:\Users\User\Downloads\toktickit> git status
● On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
● PS C:\Users\User\Downloads\toktickit> cd client
● PS C:\Users\User\Downloads\toktickit\client> npm test

> toktickit-client@1.0.0 test
> vitest run

 RUN  v2.1.9 C:/Users/User/Downloads/toktickit/client

 ✓ tests/lab-01/App.test.tsx (3)
   ✓ App (3)
     ✓ renders the TokTickIT heading
     ✓ shows Online and the seeded categories on success
     ✓ shows an Offline error message when the API is unavailable

Test Files  1 passed (1)
Tests       3 passed (3)
Start at    02:46:50
Duration    1.90s (transform 89ms, setup 161ms, collect 262ms, tests 252ms, environment 836ms, prepare 149ms)
```

figure 25 : Part 2 test evidence of client

2. docs/lab-01/tests.md

Lab 1 — Test Plan and Evidence (fill this in)

All test files live under server/tests/lab-01/ and client/tests/lab-01/.

#	Tool	Test	Result
1	Supertest	GET /api/health returns 200, status=ok	PASS
2	Supertest	GET /api/categories returns 4 seeded categories in id order	PASS
3	Vitest	Heading renders	PASS
4	Vitest	Success state shows Online + category list	PASS
5	Vitest	Error state shows Offline + message	PASS

Paste your passing terminal output / screenshot below.

✓ tests/lab-01/categories.test.ts (1) ✓ tests/lab-01/health.test.ts (1)

Test Files 2 passed (2) Tests 2 passed (2) Start at 23:48:28 Duration 668ms (transform 55ms, setup 0ms, collect 444ms, tests 69ms, environment 0ms, prepare 271ms)

✓ tests/lab-01/App.test.tsx (3) ✓ App (3) ✓ renders the TokTickT heading ✓ shows Online and the seeded categories on success ✓ shows an Offline error message when the API is unavailable

Test Files 1 passed (1) Tests 3 passed (3) Start at 23:54:10 Duration 1.51s (transform 54ms, setup 97ms, collect 188ms, tests 256ms, environment 542ms, prepare 149ms)

figure 26 : Part 2 test plan and evidence

Answer Part 3: AI Use and Reflection

1. docs/lab-01/ai_use.md

Lab 1 — AI Use and Reflection

LLM/agent used: Gemini - Flash

Selected key prompts (6–10)

#	Prompt (summarised)	What I did with the result
1	Asked for a clean Git setup process and workflow for Lab 1.	Executed step-by-step commands to initialize the repository, branches, and PRs.
2	Inquired about Error: P1001: Can't reach database server at localhost:5432 during Prisma migration.	Identified that the PostgreSQL service was not running, then started it and adjusted database configuration.
3	Asked how to resolve docker compose up -d failing with no configuration file provided.	Installed and ran PostgreSQL locally (instead of via Docker) and updated DATABASE_URL in .env to match.
4	Asked for the correct Category schema based on assignment specs.	Defined the Category model with id (Int, autoincrement), name (String, unique), and createdAt.
5	Inquired how to solve @prisma/client did not initialize yet error.	Ran npx prisma generate before migrating and seeding the database.
6	Asked for the implementation of seed.ts using upsert for the 4 categories.	Added idempotent seeding logic for "Account and Access", "Hardware", "Software", and "Network".
7	Asked how to implement the unit test in categories.test.ts.	Replaced describe.todo with assertions checking status 200, length of 4, and ordered names using Supertest.
8	Asked for the implementation of the GET /api/categories endpoint.	Added the route handler querying prisma.category.findMany ordered by id: "asc" with proper safe error handling.

Reflection

Structuring prompts by providing exact error logs and file templates helped the agent deliver accurate and immediately usable code snippets. One instance where I had to correct the agent was when it initially suggested a string UUID schema, so I provided the starter code comments to enforce PostgreSQL with integer autoincrement keys instead.

figure 27 : Part 3 ai_use.md

Answer Part 4: App Demo

1. Initial Screen

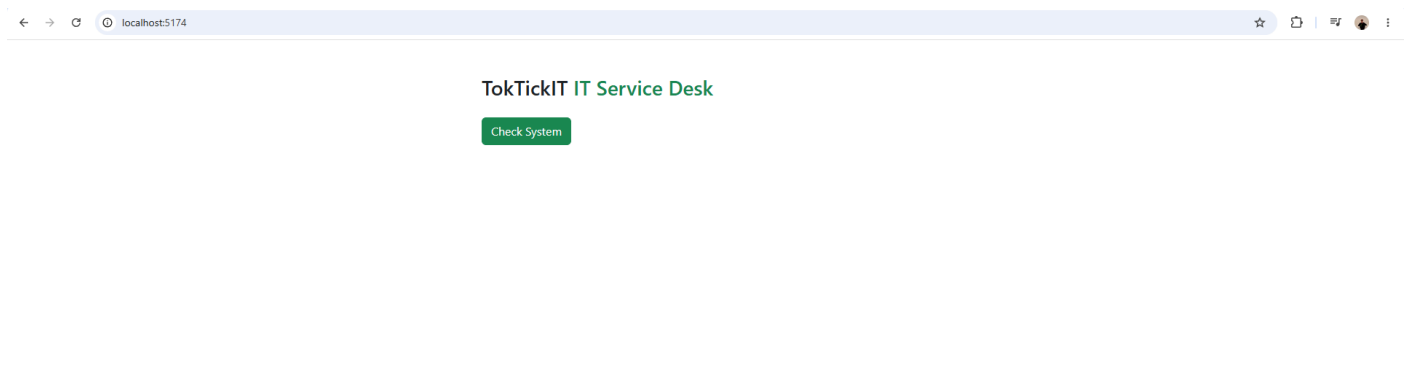


figure 28 : Part 3 Initial Screen

2. Success Case

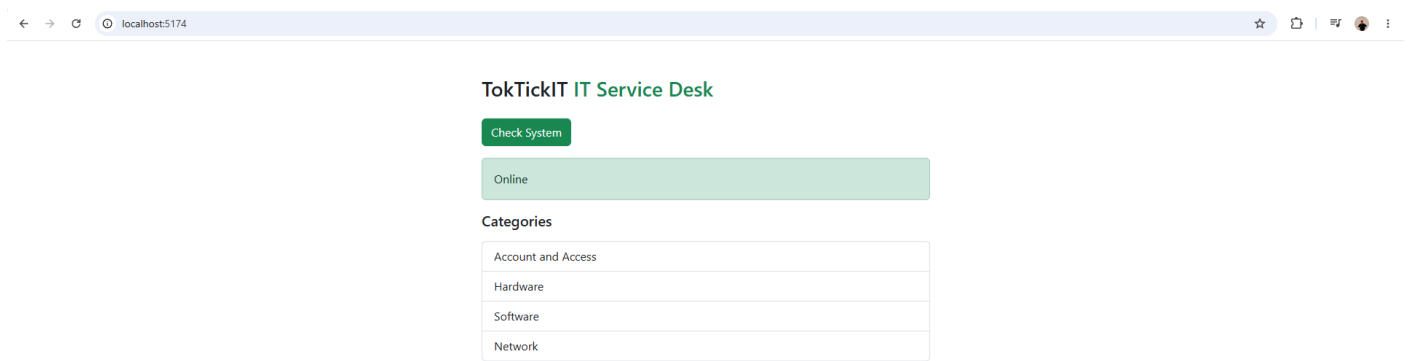


figure 29 : Part 3 Success Case

3. Failure Case

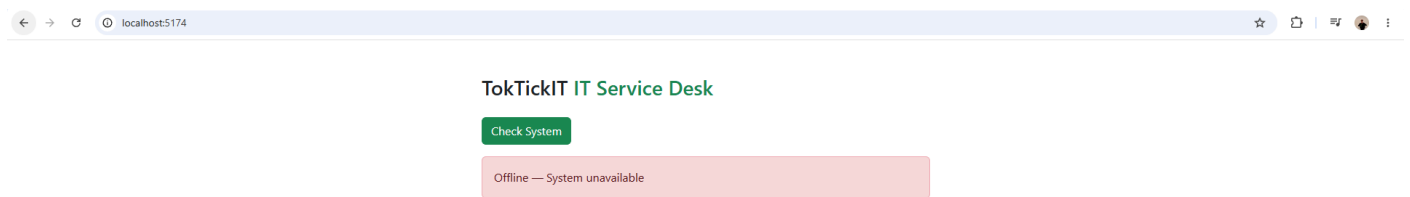


figure 30 : Part 3 Failure Case